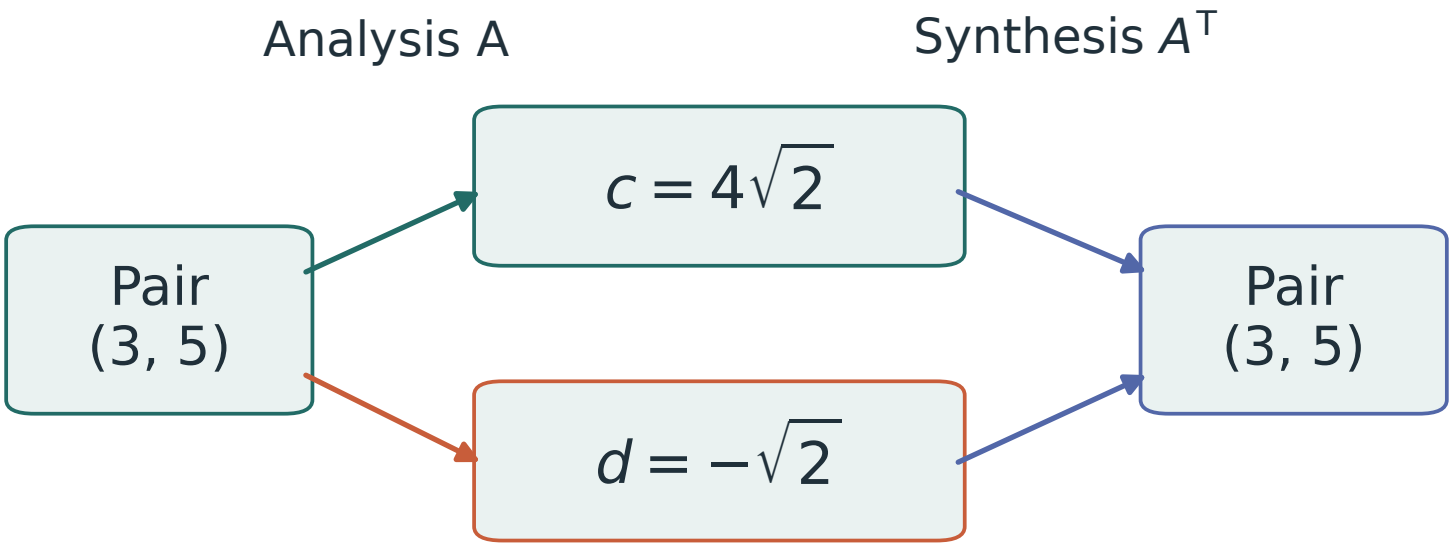

$$c = (3+5)/\sqrt{2}, \quad d = (3-5)/\sqrt{2}$$